

Lesson 6

The F test as a comparison of residual sums of squares, and restricted least squares

1 The F statistic (a reminder)

With X fixed and

$$y = X\beta + u, \quad H_0 : \underbrace{R}_{q \times K} \underbrace{\beta}_{K \times 1} = \underbrace{r}_{q \times 1},$$

Lesson 5 gave the two independent χ^2 pieces

$$\frac{1}{\sigma^2} (Rb - r)' [R(X'X)^{-1}R']^{-1} (Rb - r) \sim \chi_q^2, \quad \frac{e'e}{n - K} = s^2 \implies \frac{e'e}{\sigma^2} = \frac{1}{\sigma^2} u'Mu \sim \chi_{n-K}^2,$$

and hence

$$F(q, n - K) = \frac{(Rb - r)' [R(X'X)^{-1}R']^{-1} (Rb - r)/q}{e'e/(n - K)}.$$

2 Example: testing a subset of coefficients

Partition the regressors,

$$y = X_1\beta_1 + X_2\beta_2 + u, \quad H_0 : \beta_1 = 0,$$

with $K = k_1 + k_2$ and $q = k_1$. The restriction matrix picks out the first k_1 coefficients:

$$R = \left[I_{k_1} \mid \underbrace{0}_{k_1 \times k_2} \right] \left\{ k_1 \text{ rows} \right\}, \quad R\beta = \left[\begin{array}{cccc|cccc} 1 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \vdots & & & \vdots \\ \vdots & & \ddots & \vdots & & & & \\ 0 & 0 & \cdots & 1 & 0 & 0 & \cdots & 0 \end{array} \right] \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_K \end{bmatrix}.$$

The statistic through FWL

By Frisch–Waugh–Lovell (Lesson 4),

$$b_1 = (X_1' M_2 X_1)^{-1} X_1' M_2 y, \quad b_1 \sim N(0, \sigma^2 (X_1' M_2 X_1)^{-1}) \text{ under } H_0,$$

so

$$F(k_1, n - K) = \frac{b_1' [X_1' M_2 X_1] b_1 / k_1}{e'e / (n - K)}.$$

The numerator as a difference of sums of squares

Write the fitted regression with its residual $e = My$,

$$y = X_1 b_1 + X_2 b_2 + My.$$

Premultiplying by M_2 (which annihilates X_2 , and $M_2M = M$),

$$M_2y = M_2X_1b_1 + \underbrace{M_2My}_{My} \implies y'M_2y = b_1'X_1'M_2X_1b_1 + y'My,$$

the cross term vanishing because $X_1'M_2My = X_1'My = 0$. Hence

$$b_1'X_1'M_2X_1b_1 = y'M_2y - y'My,$$

and

$$F(k_1, n - K) = \frac{(y'M_2y - y'My)/k_1}{y'My/(n - K)}.$$

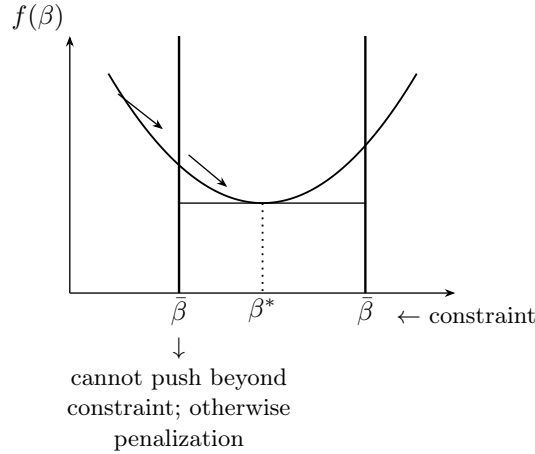
Here M_2y is the residual vector of the regression with $H_0 : \beta_1 = 0$ imposed (only X_2), and My that of the full regression. So

$$F(k_1, n - K) = \frac{(\text{RSSR} - \text{USSR})/k_1}{\text{USSR}/(n - K)},$$

with RSSR and USSR the restricted and unrestricted residual sums of squares.

3 Restricted least squares

Digression: minimising under a constraint



The minimiser β^* of $f(\beta)$ is searched for only between the two constraint lines ($\beta < \bar{\beta}$); going past a constraint is not allowed, otherwise it is penalised. The Lagrange multiplier below plays the role of that penalty.

The problem

$$\min_{b_*, \lambda_*} e_*' e_* \quad \text{s.t.} \quad Rb_* = r,$$

with Lagrangian

$$\min \mathcal{L} = (y - Xb_*)'(y - Xb_*) - 2(Rb_* - r)' \underbrace{\lambda_*}_{q \times 1},$$

λ_* being the estimate of the multiplier. Expanding,

$$\mathcal{L} = y'y - b'_*X'y - y'Xb_* + b'_*X'Xb_* - 2(Rb_* - r)'\lambda_*.$$

First-order conditions

$$\frac{\partial \mathcal{L}}{\partial b_*} = -2X'y + 2X'Xb_* - 2R'\lambda_* = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda_*} : Rb_* - r = 0.$$

The first gives

$$-X'y + X'Xb_* = R'\lambda_*.$$

Premultiply by $R(X'X)^{-1}$:

$$\begin{aligned} -R(X'X)^{-1}X'y + R(X'X)^{-1}X'Xb_* &= R(X'X)^{-1}R'\lambda_* \\ -Rb + Rb_* &= (R(X'X)^{-1}R')\lambda_* \\ -Rb + r &= R(X'X)^{-1}R'\lambda_* \quad (\text{using } Rb_* = r) \\ \lambda_* &= [R(X'X)^{-1}R']^{-1}(r - Rb). \end{aligned}$$

Substitute back λ_*

$$\begin{aligned} -X'y + X'Xb_* &= R'[R(X'X)^{-1}R']^{-1}(r - Rb) \\ -\underbrace{(X'X)^{-1}X'y}_{b} + b_* &= (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(r - Rb) \end{aligned}$$

so

$$\begin{aligned} b_* &= b + (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(r - Rb) \\ &= b - (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(Rb - r). \end{aligned}$$

The restricted residuals

$$\begin{aligned} e_* &= y - Xb_* + Xb - Xb \\ &= y - Xb - X(b_* - b) \\ &= e - X(b_* - b), \end{aligned}$$

and, since $X'e = 0$ kills the cross terms,

$$e'_*e_* = e'e + (b_* - b)'X'X(b_* - b).$$

The F statistic again

Substituting $b_* - b$,

$$\begin{aligned} e'_*e_* - e'e &= (r - Rb)'[R(X'X)^{-1}R']^{-1}R(X'X)^{-1}X'X(X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(r - Rb) \\ &= (Rb - r)'[R(X'X)^{-1}R']^{-1}(Rb - r), \end{aligned}$$

which is exactly the numerator of

$$F(q, n - K) = \frac{\overbrace{(Rb - r)' [R(X'X)^{-1}R']^{-1} (Rb - r)}^{e'_*e_* - e'e} / q}{e'e / (n - K)}.$$

Then

$$F(q, n - K) = \frac{(e'_*e_* - e'e)/q}{e'e/(n - K)} = \frac{(\text{RSSR} - \text{USSR})/q}{\text{USSR}/(n - K)}.$$

4 Properties of the restricted estimator b_*

Mean

Let

$$b_* = b - \underbrace{(X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}}_A (Rb - r) = b - ARb + Ar.$$

With $b = \beta + (X'X)^{-1}X'u$,

$$\begin{aligned} b_* &= \beta + (X'X)^{-1}X'u - AR(\beta + (X'X)^{-1}X'u) + Ar \\ &= \beta + (X'X)^{-1}X'u - AR\beta - AR(X'X)^{-1}X'u + Ar, \end{aligned}$$

so

$$\begin{aligned} \mathbb{E}(b_*) &= \beta - AR\beta + Ar \\ &= \beta \quad \text{if } H_0 (R\beta = r) \text{ holds; otherwise biased.} \end{aligned}$$

Variance

Under H_0 ,

$$\begin{aligned} \text{Var}(b_*) &= \mathbb{E}[(b_* - \beta)(b_* - \beta)'] \\ &= \mathbb{E}\left[\left((X'X)^{-1}X'u - AR(X'X)^{-1}X'u\right)\left(u'X(X'X)^{-1} - u'X(X'X)^{-1}R'A'\right)\right] \\ &= \mathbb{E}\left[(X'X)^{-1}X'uu'X(X'X)^{-1} - (X'X)^{-1}X'uu'X(X'X)^{-1}R'A' \right. \\ &\quad \left. - AR(X'X)^{-1}X'uu'X(X'X)^{-1} + AR(X'X)^{-1}X'uu'X(X'X)^{-1}R'A'\right] \\ &= \underbrace{\sigma^2(X'X)^{-1}}_{\text{Var}(b)} - \underbrace{\sigma^2(X'X)^{-1}R'A'}_{\textcircled{1}} - \underbrace{\sigma^2AR(X'X)^{-1}}_{\textcircled{2}} + \underbrace{\sigma^2AR(X'X)^{-1}R'A'}_{\textcircled{3}}. \end{aligned}$$

①, ② and ③ are the same; each equals

$$\sigma^2(X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}R(X'X)^{-1}$$

(for ③, $R(X'X)^{-1}R'$ cancels against the inverse inside A). Hence

$$\text{Var}(b_*) = \text{Var}(b) - \underbrace{\sigma^2(X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}R(X'X)^{-1}}_{\text{p.s.d. matrix}},$$

so b_* is even more efficient than b .

Without H_0

In general

$$b_* = \beta - AR\beta + Ar + (X'X)^{-1}X'u - AR(X'X)^{-1}X'u,$$

and the variance is taken about the mean,

$$\text{Var}(b_*) = \mathbb{E}\left[(b_* - \mathbb{E}(b_*))(b_* - \mathbb{E}(b_*))'\right].$$

When is b_* biased and when unbiased? (added discussion)

The note at the end of the page asks for this; it is not part of the lecture.

- **The bias.** From the mean above,

$$\mathbb{E}(b_*) - \beta = -A(R\beta - r) = -(X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(R\beta - r).$$

A is $K \times q$ of full column rank q (because R has rank q), so $A\delta = 0$ only for $\delta = 0$. Hence

$$b_* \text{ is unbiased} \iff R\beta = r,$$

i.e. exactly when the restriction imposed is true. When it is false, the bias is proportional to how badly it fails, $R\beta - r$.

- **The variance does not depend on H_0 .** Whether or not $R\beta = r$,

$$b_* - \mathbb{E}(b_*) = (I - AR)(X'X)^{-1}X'u,$$

because the terms $\beta - AR\beta + Ar$ are non-random and drop out. So the variance computed under H_0 , $\text{Var}(b_*) = \sigma^2(X'X)^{-1} - \sigma^2(X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}R(X'X)^{-1}$, is the variance of b_* in every case. A false restriction moves the centre of b_* , not its spread.

- **No conflict with Gauss–Markov.** b is best among linear *unbiased* estimators for every β . b_* beats it only by using extra information, the restriction; if that information is wrong, b_* is no longer unbiased and falls outside the Gauss–Markov class.
- **A bias–variance trade-off.** With a false restriction, compare the two by mean squared error,

$$\begin{aligned} \text{MSE}(b_*) &= \text{Var}(b_*) + (\mathbb{E}b_* - \beta)(\mathbb{E}b_* - \beta)' \\ &= \text{Var}(b) - \sigma^2(X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}R(X'X)^{-1} \\ &\quad + A(R\beta - r)(R\beta - r)'A'. \end{aligned}$$

If the restriction is only slightly wrong, the variance gain can outweigh the squared bias and b_* may still be preferred; if it is badly wrong, the bias dominates.

- **The link to the F test.** Whether the restriction is true is exactly what $H_0 : R\beta = r$ tests. A small $F = \frac{(\text{RSSR} - \text{USSR})/q}{\text{USSR}/(n - K)}$ means imposing the restriction barely raises the residual sum of squares, which is evidence that b_* is (close to) unbiased; a large F rejects H_0 and warns that b_* is biased.
- **The same for s^2 .** The restricted variance estimator $e'_*e_*/(n - K + q)$ is unbiased for σ^2 under H_0 ; when $R\beta \neq r$, e'_*e_* contains the extra $(Rb - r)'[R(X'X)^{-1}R']^{-1}(Rb - r)$ term with a non-zero mean, and it overstates σ^2 .

Transcription notes

- The set opens with a divider page carrying only “Lesson 6”; it is not reproduced.
- As in Lessons 3–5 the regressor matrix, written lower case (x) on the page, is typeset X , and B for β is typeset β . The page writes the restricted estimator both b_* and b^* and the multiplier both λ_* and λ^* ; typeset b_* and λ_* throughout. K and k are both used for the number of regressors; typeset K .
- Page 2: the first χ^2 is written “ $\sim \chi_q^2$ ”; typeset χ_q^2 . The second line reads “ $e'e/(n-k) = s^2 \Rightarrow e'e/(n-k) \frac{1}{\sigma^2} u'Mu \sim \chi^2$ ”; typeset as $e'e/\sigma^2 = \frac{1}{\sigma^2} u'Mu \sim \chi_{n-K}^2$ (the repeated $e'e/(n-k)$ is read as a slip for $e'e/\sigma^2$; the degrees of freedom are **added**). **Corrected:** the F numerator on page 2 is written $(R(X'X)^{-1}R')$ without the inverse; page 6 has it with the inverse.
- Page 2: R is drawn as $[I_{k_1} \mid 0_{k_1 \times k_2}]$ with a brace “ k_1 ” and then as an explicit $1/0$ block matrix; both kept.
- Page 3: “ $\sim N(0, \dots)$ ” for b_1 holds under H_0 ; “under H_0 ” is **added**. The numerator of the first $F(k_1, n-K)$ in y ’s is written “ $(y'M_2y - y'My)k_1$ ”; typeset $/k_1$, as in the RSSR line below it. “ $H_0 : \beta = 0$ ” beside M_2y and M_y is typeset $\beta_1 = 0$. The reasons $M_2M = M$ and the vanishing cross term are **added**.
- Page 4, the figure: a convex $f(\beta)$, two vertical constraint lines both labelled $\bar{\beta}$ on the page (kept), the minimum β^* between them marked by a horizontal tangent and a dotted drop line, two small arrows down the left branch of the curve, the label “constraint” at the right line and “cannot push beyond constraint. otherwise penalization” under the left. The sentence after the figure is **added** to tie it to the multiplier. “ $\beta < \bar{\beta}$ ” is written beside it and kept in that sentence.
- Page 4: the dimension written under the Lagrangian reads “ $K \times q$ ” under R and “ $q \times 1$ ” under λ_* ; only the latter is kept (R is $q \times K$). The expansion of $\mathcal{L}$ on the page stops at “ $-(Rb_* - r)'$ ”; completed as $-2(Rb_* - r)'\lambda_*$. **Corrected:** the FOC is written $-2X'y + 2X'Xb_* + 2R'\lambda_* = 0$; with $-2(Rb_* - r)'\lambda_*$ in $\mathcal{L}$ the last term is $-2R'\lambda_*$, which is what the page’s next lines (with $R'\lambda_*$ on the right-hand side) use. **Corrected:** the next line is written $-X'y + X'X + 2R'\lambda_* = 0$; typeset $-X'y + X'Xb_* = R'\lambda_*$. An arrow on the page links $Rb_* - r = 0$ to the step $-Rb + r$.
- Page 5: **Corrected:** b_* is written with $(X'X)R'$ (no inverse) in all three lines; typeset $(X'X)^{-1}R'$, as on page 7. The arrow from $(X'X)^{-1}X'y$ to b is typeset as an underbrace. “Substitute” spelling fixed. The reason the cross terms in e'_*e_* vanish ($X'e = 0$) is **added**.
- Page 6: **Corrected:** the left-hand side is written e'_*e_* ; the quadratic form is $e'_*e_* - e'e$ (page 5, last line), as the F line below it uses. The first bracket is written $[RX'X)^{-1}R']^{-1}$; typeset $[R(X'X)^{-1}R']^{-1}$. The underline under the F numerator is typeset as an overbrace labelled $e'_*e_* - e'e$.
- Page 7: **Corrected:** b_* is written $b + A(Rb - r)$; page 5 and the expansion directly below (with $-AR(\dots) + Ar$) give $b - A(Rb - r)$. “ H_0 under $Rb = r$ ” is typeset $R\beta = r$; “if $H_0 \checkmark$ ” as “if H_0 holds”. **Corrected:** the second factor of the variance is written $(u'X(X'X)^{-1} - u'(X'X)^{-1}R'A')$; the X after u' is restored. “Under H_0 ” before the variance is **added** (the page centres at β).
- Page 8: ①, ②, ③ are the page’s own labels; the line explaining why they are equal is **added**. **Corrected:** the page writes “ $\text{Var}(b_*) = \text{Var}(b) - \text{p.d. matrix}$ ”; the matrix has rank $q \leq K$, so it is positive *semi*-definite (p.d. only when $q = K$); typeset p.s.d. “efficient” spelling fixed.
- Page 9 carries only $\text{Var}(b_*) = \mathbb{E}[(b_* - \mathbb{E}b_*)(b_* - \mathbb{E}b_*)']$ and the note “Claude: Add discussion on when b_* biased and unbiased”. The whole subsection “When is b_* biased and when unbiased?” is **added** in answer to it and is not on the page.
- Pages of the original, for tracing back: page 1 the divider; page 2 the model, H_0 , the two χ^2 pieces, $F(q, n-K)$, and the subset example with R ; page 3 b_1 by FWL, its F , the M_2y decomposition and the RSSR/USSR form; page 4 the constraint figure, the Lagrangian, the FOC and λ_* ; page 5 substituting λ_* back, b_* , e_* and e'_*e_* ; page 6 F , $e'_*e_* - e'e$ and the RSSR/USSR form; page 7 A , the expansion of b_* , $\mathbb{E}(b_*)$ and the start of $\text{Var}(b_*)$; page 8 the four variance terms, ①–③, $\text{Var}(b_*) = \text{Var}(b) - \text{p.s.d.}$, and b_* without H_0 ; page 9 the general variance and the note.